Expectation, Variance, and Discrete Distributions

50 minutes • 50 points • Student quiz

Name: _____ **Date:** _____

Instructions. Answer all four problems and show your reasoning. No calculator is needed; exact fractions or exact expressions with powers and binomial coefficients are acceptable. You may use the standard binomial and geometric mean and variance formulas without proof. Geometric waiting times count trials *including the first success*. The suggested times leave 4 minutes to check your work.

1. From outcomes to moments

14 points · about 13 minutes

Let $\Omega = \{a, b, c, d\}$, with every subset an event. The outcome probabilities and the values of a random variable X are given below; q is a constant.

Outcome ω	a	b	c	d
$\mathbb{P}(\{\omega\})$	q	$2q$	q	$4q$
$X(\omega)$	-2	0	2	2

(a) Find q. List the outcomes in the event $\{X > 0\}$ and find $\mathbb{P}(X > 0)$. (3 points)

(b) Give the probability mass function of X: list each distinct possible value and its probability. (3 points)

(c) Compute $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, and $\mathrm{Var}(X)$. (5 points)

(d) Let $Y = 3 - 2X$. Find $\mathbb{E}[Y]$ and $\mathrm{Var}(Y)$, stating the transformation rules you use. (3 points)

2. Bernoulli indicators

10 points · about 9 minutes

Roll a fair six-sided die once. Let $I = 1$ if the result is 5 or 6, and let $I = 0$ otherwise.

(a) State the distribution of I, including its parameter, and give $\mathbb{P}(I = 0)$ and $\mathbb{P}(I = 1)$. (2 points)

(b) Use the two possible values of I to calculate $\mathbb{E}[I]$, $\mathbb{E}[I^2]$, and $\mathrm{Var}(I)$. Show the weighted sums rather than only quoting a Bernoulli formula. (4 points)

(c) On the *same roll*, let $J = 1 - I$. Find $\mathbb{E}[I + J]$ and $\mathrm{Var}(I + J)$. A student claims $\mathrm{Var}(I + J) = \mathrm{Var}(I) + \mathrm{Var}(J) = 4/9$. Explain why this reasoning fails here. (4 points)

3. Binomial counts

12 points · about 11 minutes

A bag contains one red ball and three blue balls. Draw one ball uniformly at random, record its color, and replace it. Repeat for a total of six draws. Assume the draws are independent. Let N be the number of red balls recorded.

(a) State the distribution of N, its parameters, and its possible values. Explain why the model applies. (3 points)

(b) Find $\mathbb{P}(N = 2)$. Explain the counting factor in your expression. (3 points)

(c) Find the probability that at least one red ball is recorded. (2 points)

(d) Compute $\mathbb{E}[N]$ and $\mathrm{Var}(N)$. If N is written as a sum of six success indicators, why may their variances be added? (4 points)

4. Geometric waiting times

14 points · about 13 minutes

A device attempts to send a message repeatedly. Each attempt succeeds with probability 1/4, independently of all other attempts. Let T be the number of attempts *up to and including the first success*.

(a) Give the possible values of T and a formula for $\mathbb{P}(T = k)$. Explain the formula and use a geometric series to check that its probabilities sum to 1. (4 points)

(b) Find $\mathbb{P}(T = 3)$ and $\mathbb{P}(T > 3)$. (2 points)

(c) Find $\mathbb{E}[T]$ and $\mathrm{Var}(T)$. (3 points)

(d) Let F be the number of *failed* attempts before the first success. Express F in terms of T, then find $\mathbb{E}[F]$ and $\mathrm{Var}(F)$. (2 points)

(e) Given that the first two attempts have failed, find the probability that more than five attempts in total will be needed: $\mathbb{P}(T > 5 \mid T > 2)$. Justify your answer. (3 points)